

SELF-CORRECTIVE MULTIMODAL RAG

python · fastapi · langgraph · postgres/pgvector · redis · ollama

Retrieval that grades itself, and refuses when the evidence isn't there.

Prism runs every question through a LangGraph correction loop: retrieve, grade, rewrite, retrieve again, rerank, generate, then verify the answer against the chunks that produced it. The loop runs on a local GPU by default and escalates to a paid model only when it has to, so most queries cost nothing. Text and figures are indexed together, every claim carries a citation you can click back to its source region, and answers below the groundedness threshold return an abstention instead of a guess.

Open the query console

See a live trace

runs on one machine with a consumer GPU

94%

of calls served on a local GPU at \$0

\$0.00019

mean cost per query, all-in

2.71s

p95 end to end, including retries

1 / 72

injection payloads that still succeed

The correction loop

one graph per query · every stage graded, logged, replayable

01

plan_query

Decompose the question, detect the likely modality, and pick the collections in scope.
qwen2.5:14b · ollama local · ~210ms

02

retrieve

Hybrid BM25 + HNSW over pgvector. Text and figure chunks share one index and one score floor.
pgvector · redis cache · ~70ms

03

grade_docs

A local grader judges whether the retrieved set can actually support an answer, and says why not. Free, so it can run on every attempt.
llama3.1:8b · ollama local · ~280ms

04

rewrite_query

On a failed verdict the question is rewritten with the grader's reasoning, then retrieval runs again. Up to three attempts.
qwen2.5:14b · ollama local · ~360ms

05

generate

Answer written against reranked chunks only, with every claim bound to a chunk span. Local model first; escalates to Gemini or OpenAI under a per-query cost budget.
qwen2.5:32b local → gemini → openai

06

verify_grounding

The answer is re-checked against its own citations. Below it becomes an abstention, not a hedge.
llama3.1:8b · ollama local · ~240ms

What operators get

six surfaces, one telemetry model

Streamed answers with clickable provenance

Every citation opens the exact chunk that produced it — the highlighted sentence for text, the parsed bounding box for a table or figure.

- live correction-loop side panel
- per-stage model, verdict and latency
- abstention as a designed state

The whole graph execution, node by node

One query as a LangGraph run: model, latency, token cost and pass/fail per node, with rejected retrieval branches kept visible but dimmed.

- latency and cost waterfall
- node output payload inspector
- fork from any node with new parameters

What correction actually costs

p50/p95/p99 latency, cost per query, cache hit rate and call mix over time, next to a paid-API single-shot baseline.

- retry distribution across the loop
- self-hosted carries the calls, not the spend
- ~95% cost and ~34% p95 vs baseline

QUEUE & SCALING

Concurrency on one GPU

Queue depth with the autoscaling threshold marked, worker count overlaid on depth, and throughput against time spent queued versus executing.

- autoscaler keyed on depth, not CPU
- Ollama Cloud pinned at its cap of 1
- load test to 64 concurrent requests

COLLECTIONS & KEYS

Scoped access and ingestion state

Collections with live ingestion progress and embedder per collection; API keys scoped to collections with per-key usage and quota.

- per-collection cache scope and hit rate
- quarantine for documents that fail OCR
- per-key abstention threshold override

EVAL REGRESSION

Golden-set metrics per commit

Accuracy, groundedness and refusal rate trended across commits, with the regressing commit flagged and its worst slice named.

- breakdown across the four serving providers
- slice-level diagnosis, not just a number
- runs on commit · 30-query golden set

ROBUSTNESS

Indirect prompt-injection results

Attack success rate before and after mitigations, by category and vector, reported as a security document rather than a marketing claim.

- 72 payloads, five categories
- mitigation cost in ms and dollars
- open findings listed with severity

Where it earns its keep

corpora where a confident wrong answer costs more than no answer

Financial filings and disclosures

Answer questions across 10-Qs and policy notes where the number lives in a table, not a sentence.

multimodal audit trail
0.94 groundedness on the table-heavy slice

Regulated support and claims

Agents get an answer with its source, or an explicit abstention they can escalate — never an invented policy.

abstention citations
0% answers without a bound citation

Engineering knowledge bases

Wikis and design docs synced on push, where stale and current answers look identical without provenance.

git sync freshness
1,340 chunks kept current per repo

Scanned document sets

Quarantine for documents that fail OCR, so a silent coverage gap never becomes a confident answer.

quarantine scoped keys
94% corpus indexed, remainder flagged

Scientific and technical literature

Local-first cost engineering

Figure captions, plots and equations are indexed with their surrounding text and cited as regions.

figures regions

2 of 8 chunks in a typical answer are figures

Route each graph stage to a local or paid model and compare accuracy, cost and latency on your own golden set.

4 serving providers fork & replay

+7 / 30 correct answers over single-shot RAG

Abstention is a feature

A tunable groundedness threshold decides when Prism answers. Below it, the response is an abstention that names the coverage gap, the best chunk it found, and what to ingest to close it — the same payload your application can render as a first-class state.

ABSTAINED groundedness 0.41 < τ 0.58

No indexed document supports a claim about FY27 guidance. Closest match discusses FY26 only.

Hardened against injected context

Retrieved text never enters the instruction frame. Figure captions are sanitized after OCR, cited chunk ids are verified against the retrieved set, and generation nodes have no network egress. The suite runs 72 indirect-injection payloads across five categories on every release.

payloads still succeeding confirmed exfiltration paths open findings p95 mitigation overhead

1 / 72 0 1 +118ms

Read the security report →

Runs on your infrastructure

no vendor-managed vector store, no proprietary format

ORCHESTRATION

LangGraph

explicit nodes and edges; every run persisted as a trace

API

FastAPI · Python 3.12

streaming SSE, OpenAPI schema, scoped key auth

VECTOR STORE

Postgres 16 + pgvector

HNSW, hybrid BM25, your database and your backups

CACHE & QUEUE

Redis

per-collection cache scope, ingestion queue, autoscaling signal

MODELS

local first, per node

ollama self-hosted → ollama cloud → gemini → openai

FRONTEND

Next.js · shadcn/ui · Recharts

this console, embeddable per surface

Point it at one collection and watch the loop work.

Docker compose up, pull the Ollama models, ingest a folder, open the trace viewer. No API key needed until the router escalates.

Start ingesting

See the benchmarks



Operator system

Charts are hand-built from the same ramp — no default chart palette appears anywhere in the product.

```
PROVIDER SERIES
Fixed provider—hue map. A provider keeps its hue on
every chart on every screen.

ollama · self-hosted · #4F8CFF
ollama cloud · free tier · #7AA2F7
geminai · #B07CF0
openai · #E0748F
in-process · pgvector, rerank ·
#4D5A63
naive baseline · dashed, never filled
```

TYPE SCALE

Screen title 19/600/-2%

Card heading 13/600

Body / UI default 12/400

248 ms · **\$0.0041** mono 12/500 · metric

SECTION LABEL mono 9.5/500/+10%

1.84_s mono 24 · stat

>

What did the Q3 filing say about deferred revenue recognition for multi-year contracts?

Low-evidence example

SCOPE

sec-filings

axxiv-sample

τ

ABSTAIN

0.58

MAX RETRIES

3

ROUTER

local-first cascade

COST BUDGET

\$ 0.0050

LATENCY BUDGET

6

s

ANSWER

complete · 2.76s · verified 0.94

Multi-year contracts are recognized ratably over the contract term, with the unearned portion held in deferred revenue and reported in two buckets, current and long-term. [1](#)

At quarter end the current portion was \$184.2M against \$96.7M long-term, a 12% sequential increase driven by three enterprise renewals signed in the final two weeks of the period. [2](#) [3](#)

The filing notes one policy change: contracts with usage-based overage components now defer the entire overage estimate until the annual true-up, rather than accruing monthly. [4](#)

GROUNDING 0.94

\$0.0018 of \$0.0050 · 2.76s of 6s

WITHIN BUDGET

4 citations · 4 chunks · 0 unsupported spans

Open trace →

SOURCE CHUNKrerank 0.94 · vec 0.91

TEXTsec-filings/FY26-Q3-10Q.pdf · p.28 \$revenue

For arrangements spanning more than twelve months, the Company recognizes revenue [ratably over the contract term](#), with amounts not yet earned presented as deferred revenue and disaggregated between [current and long-term portions](#). Contract modifications are accounted for prospectively unless the modification adds distinct goods or services.

- 1 · 10-Q p.28
- 2 · Table 4 · figure
- 3 · 10-Q p.29
- 4 · policy note p.44

CORRECTION	LOOP	retry 1/3
ELAPSED	COST	
2.76s	\$0.0018	
CHUNKS KEPT	TOKENS	
8/24	14.2k	
plan_query OK	qwen2.5:14b · ollama 210ms	Split into two sub-questions; tagged the answer as likely table-grounded. Routed local.
retrieve · attempt 1 12/24	pgvector hns w k=24 61ms	Twelve chunks above the score floor; best 0.58, none from a table region.
grade_docs FAIL 0.38	llama3.1:8b · ollama 284ms	Chunks describe FY25 policy language, not multi-year deferral mechanics. Rewrite requested.
rewrite_query OK	qwen2.5:14b · ollama 361ms	Added "ratably" and "unearned"; forced hybrid retrieval on the rewritten form. Still \$0.
retrieve · attempt 2 18/32	hybrid bm25 + hns w 74ms	Six figure chunks surfaced, including Table 4 on p.29.
grade_docs PASS 0.86	llama3.1:8b · ollama 268ms	Eight chunks retained and passed to the reranker.
generate ESCALATED	gemini-2.5-flash 1090ms	Two figure chunks in context, so generation escalated off the local model. \$0.0018 of the \$0.0050 budget.
verify_grounding PASS 0.94	llama3.1:8b · ollama 242ms	No unsupported spans. Above τ 0.58, so the answer is returned rather than abstained.

trace_01J08F2K7XR ANSWERED 1 rewrite · 10 nodes · 2 pruned

2.76s | \$0.0018 | 14.2k tok

grade_docs FAIL

llama3.1:8b · ollama local

plan_query qwen2.5:14b · ollama loc... 210ms \$0.0000

embed_query nomic-embed-text · ollam... 46ms \$0.0000

retrieve · try 1 THIN pgvector hns w · k=24 61ms 12 kept

grade_docs FAIL llama3.1:8b · ollama loc... 284ms rel 0.38

web_fallback not taken · gate off

rewrite_query qwen2.5:14b · ollama loc... 361ms \$0.0000

retrieve · try 2 hybrid bm25 + hns w · k=32 74ms 18 kept

abstain not reached

grade_docs llama3.1:8b · ollama loc... 268ms rel 0.86

rerank bge-reranker-v2-m3 · inr... 112ms 8 kept

generate · escalated gemini-2.5-flash 1.09s \$0.0018

verify_grounding llama3.1:8b · ollama loc... 242ms \$0.0000

LATENCY 284ms

TOKENS IN/OUT 6.1k / 210

COST \$0.0000

CACHE miss

NODE OUTPUT

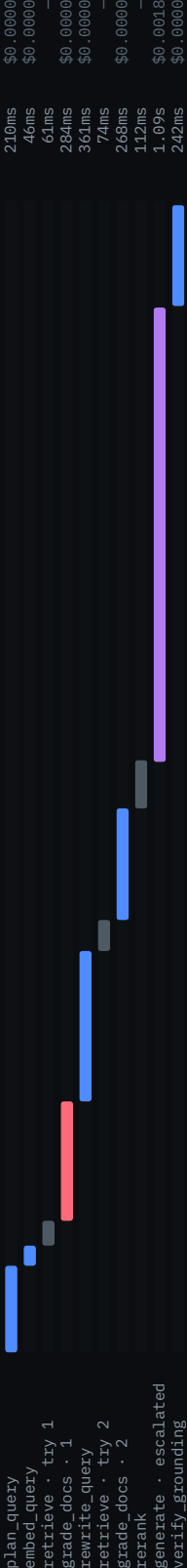
```
{
  "verdict": "insufficient",
  "relevance": 0.38,
  "reason": "chunks cover FY25 policy, not multi-year deferral mechanics",
  "action": "rewrite_query"
}
```

REPLAY

Fork the graph at this node. Upstream state is reused from the trace; everything downstream re-executes with the parameters you set.

Fork from grade_docs

LATENCY & COST WATERFALL



autoscaler keyed on queue depth, not CPU scale-up at depth > 8 sustained 10s · scale-down at depth < 2 for 60s · ceiling 8 workers

last 60 minutes

QUEUE DEPTH

3

scale-up threshold 8

ACTIVE WORKERS

4 / 8

ceiling 8 · 2 scale events in the hour

THROUGHPUT

2.4

queries / sec · peak 3.9

TIME IN QUEUE

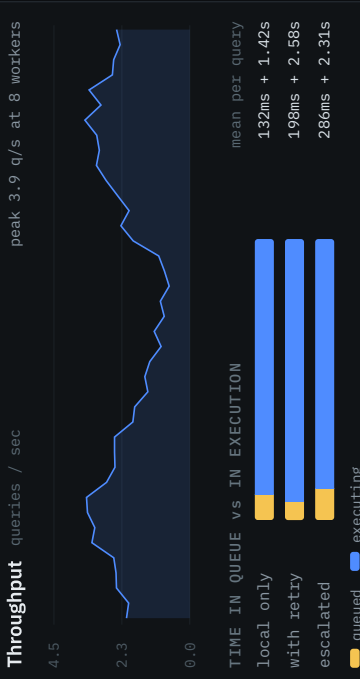
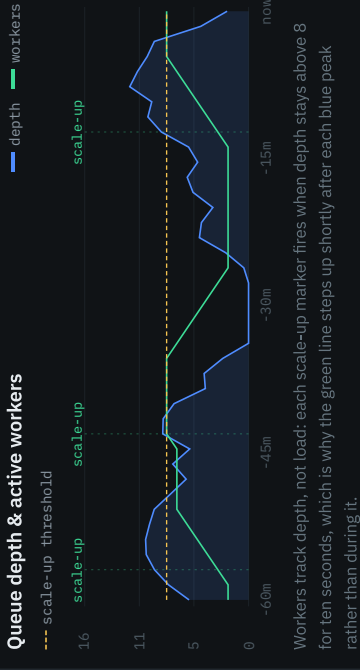
184ms

mean · 6.7% of end-to-end

REJECTED

0

no request dropped at the gate



Concurrency utilization by provider

- ollama · self-hosted**
graders, rewrite, embeddings, most generation
in flight / permitted
6 / 8
- ollama cloud · free tier**
hard concurrency 1 · GPU-time metered
1 / 1 **CAP**
- gemini**
multimodal ingestion + escalated generation
2 / 4
- openai**
final escalation, used sparingly
0 / 2
- in-process**
pgvector, rerank · bound by worker count
4 / 8

Ollama Cloud is a free tier with a hard concurrency of 1 and GPU-time metering, so it is a single-slot escalation lane and never a scaling target. Self-hosted Ollama scales with the worker pool; the paid providers are held to small explicit caps.

Load test · concurrent requests

30-query set replayed · single RTX 4090 host

CONC.	p50	p95	p99	vs BASELINE
1	1.61s	2.24s	2.48s	-38%
8	1.74s	2.71s	3.06s	-34%
32	1.98s	3.84s	5.12s	-21%
64	2.14s	6.41s	9.88s	-9%

Past 32 concurrent requests the local GPU saturates and queue time dominates: p99 rises while p50 holds, which is the signature of queueing rather than slow execution. The paid-API baseline degrades earlier because every stage pays a network round trip.

1h 24h 7d last 24 hours · 1-hour buckets · 612 queries

p95 LATENCY

2.71s

-4.4% vs prior window

COST / QUERY

\$0.00019

-95.5% paid-API baseline \$0.0042

End-to-end latency p50 / p95 / p99 · ms



Correction-loop retries



Queries exhausting all three retries end in abstention in 17 of the 20 cases seen so far — the loop is working as a filter, not a retry-until-answer machine.

Paid-API baseline vs Prism local-first

p95 LATENCY

2.71s

-34.2%

COST / QUERY

\$0.00019

-95.5% · local-first

PAID CALLS

5.8%

was 100%

94.2% served locally

ACCURACY · 30 Q

26 / 30

was 19 / 30

+7 queries

GROUNDEDNESS

0.94

was 0.80

+0.14

TOKENS / QUERY

14.2k

was 9.1k

+56% · nearly all on local GPU

ollama

ollama-cloud

gemini

openai

in-process

CACHE HIT

37.4%

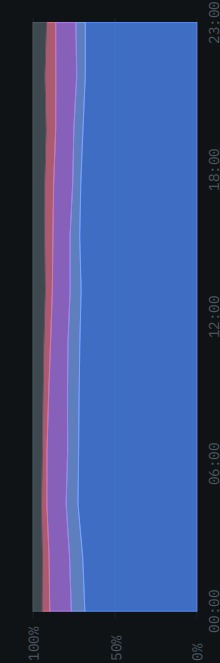
+2.1pp per-collection scope

LOCAL-SERVED

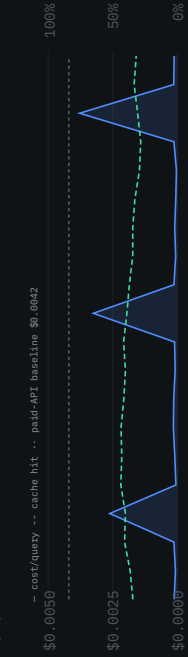
94.2%

+1.8pp ollama self-hosted

Provider mix



Cost per query & cache hit rate



golden set · 30 queries · 3 collections

runs on commit · last run 2h ago · 4m 12s

1 REGRESSION



Commits newest first · metrics vs previous commit

COMMIT	SUBJECT	ACCURACY	GROUNDING	REFUSAL	p95
9d31f7a	harden figure-caption sanitizer before rerank	0.867	0.944	10.0%	2.71s
b71c04e	scope rewritten-query cache per collection	0.867	0.943	10.0%	2.61s
2f88ad1	move grader to llama3.1:8b on local gpu	0.867	0.941	10.0%	2.74s
4e2a1c9	chunker: 512-token windows, 64 overlap	0.800	0.819	13.3%	2.94s
7c19b04	add table-aware layout parser v3	0.867	0.938	10.0%	2.81s
a03ff52	abstain when top rerank score < 0.44	0.867	0.931	10.0%	2.78s
d5610bb	gemini ingestion for figure + equation chunks	0.833	0.922	10.0%	2.85s
1ab99c7	baseline: single-shot top-k, paid api only	0.633	0.802	3.3%	4.12s

Accuracy by generator



REGRESSION 4e2a1c9 2 commits back · 2d ago

Groundedness fell 0.041 after the chunker switched to 512-token windows. The drop is concentrated in table-heavy documents: two of the six figure-grounded queries flipped to abstention while prose held flat.

worst slice
unaffected
retry rate
figure-grounded · 6/6 → 4/6
prose · 22/24 → 22/24
+6.7pp · loop absorbing the loss

Open 4 failing traces

Diff vs 7c19b04

indirect prompt-injection suite · 72 payloads · 5 categories

last full run 2026-09-07 · mitigations at commit 9d31f7a

AGGREGATE ASR

1 / 72

was 13 / 72 before mitigations

CONFIRMED EXFIL

0

no tool call reached an external host

OPEN FINDINGS

1

PRSM-441 · rendered-in-image caption text

MITIGATION COST

+118ms

p95 overhead · \$0 (all local)

Attack success rate by category

Instruction override in body text

18 payloads · text layer

Hidden text in figure captions

16 payloads · image + caption

Tool-call smuggling via markdown

12 payloads · link syntax

Retrieved-context role confusion

14 payloads · chunk framing

Citation spoofing

12 payloads · fabricated chunk id

before mitigations after

4 / 18 → 0 / 18

5 / 16 → 1 / 16

2 / 12 → 0 / 12

2 / 14 → 0 / 14

2 / 12 → 0 / 12

Aggregate ASR over releases

payloads succeeding, of 72



Active mitigations

Context/instruction separation

retrieved text never enters the system frame

+0ms

ON

Caption sanitizer v2

strips imperative spans from OCR output

+41ms

ON

Citation id verification

every cited chunk id must exist in the retrieved set

+12ms

ON

Tool-call allowlist

no network egress from generation nodes

+0ms

ON

Second-pass injection classifier

Llama3.1:8b screens the assembled context locally

+65ms

CANARY

The one payload that still succeeds is a figure caption whose injected text is rendered inside the image rather than the text layer, so the sanitizer never sees it. Tracked as PRSM-441; no confirmed exfiltration path.

Collections 3 collections · 4,180 chunks · 412 MB in pgvector

COLLECTION	DOCS	CHUNKS	EMBEDDER	CACHE SCOPE	INGESTION	STATE
sec-filings pdf · 10-Q, 10-K · table-aware parser, gemini ingestion	164	2,410	nomic-embed-text	collection-local 41% hit	100%	READY
engineering-wiki markdown · synced from git on push	212	1,340	nomic-embed-text	collection-local 33% hit	100%	READY
arxiv-sample pdf · figures and equations · 3 docs failed OCR, quarantined	48	430	nomic-embed-text	collection-local 18% hit	94%	PARTIAL

+ Ingest source

Scoped API keys cache keys are namespaced per collection – a cross-collection hit would be a leak, so it cannot occur

+ New key

demo-ui · local prism_live_...4f21 ACTIVE

eval-runner prism_test_...1b83 ACTIVE

load-test prism_test_...9c07 RATE-LIMITED

Abstention threshold

groundedness below τ returns an abstention instead of an answer

0.58 τ global default



projected abstention rate



23.8%

answer coverage



76.2%

unsupported-claim risk



8.1%

Projected from every graded trace so far, at 0.01 increments, 0.55–0.62 is where coverage and unsupported-claim risk cross on the 30-query golden set.

Apply to all keys

Per-key override